SMART BANK LOAN MANAGEMENT AND AUTOMATION SYSTEM

A PROJECT REPORT

Submitted by

HARINI NAGARAJAN

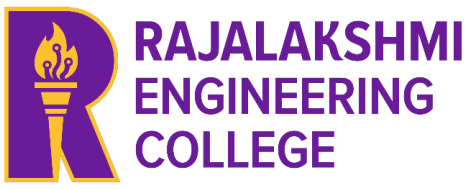
(240701167)

in partial fulfilment of the End Semester Examination for the course

CB23F35 – LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled "Smart Bank Loan Management and Automation System" is the bonafide work of HARINI NAGARAJAN (240701167), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under the guidance of Dr. Jinu Sophia.

SIGNATURE

Dr. Manoranjini J
Head of the Department
Department of Computer Science and Engineering
Rajalakshmi Engineering College

SIGNATURE

Dr. Jinu Sophia
Course Faculty
Department of Computer Science and Engineering
Rajalakshmi Engineering College

Submitted for the End Semester Examination (Individual Project) held on 10.11.2026.

INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

I, HARINI NAGARAJAN (240701167), hereby declare that the project report titled "Smart Bank Loan Management and Automation System", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of individual project work carried out by me.

The Zoho Creator application described in this report includes its forms, reports, pages, Deluge scripts, workflows, approvals, Blueprint and dashboards. The report is based on the project application structure and the implementation details available at the time of preparation.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Any external references, platform documentation, templates or AI-assisted resources used in preparing the work are acknowledged where applicable.

Place: Chennai

Date: 10.11.2026

SIGNATURE OF THE STUDENT

HARINI NAGARAJAN

240701167

ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman Thiru. S. Meganathan, Chairperson Dr. (Mrs.) Thangam Meganathan and Vice-Chairman Mr. Abhay Shankar Meganathan for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, Dr. S. N. Murugesan, Rajalakshmi Engineering College, for the encouragement extended to students to undertake industry-relevant learning.

I thank Dr. Manoranjini J, Head of the Department, Department of Computer Science and Engineering, for the guidance and support provided throughout the course.

I owe my sincere thanks to Dr. Jinu Sophia, Course Faculty, for the valuable guidance, reviews and technical feedback that supported the development of this application and report.

I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and its learning resources. I thank my family and friends for their encouragement and support.

HARINI NAGARAJAN

ABSTRACT

Retail banking requires accurate recordkeeping, efficient loan onboarding, objective document verification, credit-risk checks, controlled disbursement and transparent repayment tracking. In manual processes, customer details may be scattered across paper files and spreadsheets. Validation errors, delayed approvals, manual document checks and late identification of overdue instalments can affect service quality and operational efficiency.

This project presents the Smart Bank Loan Management and Automation System developed using Zoho Creator. The application includes ten forms—Applicant_Details, Branches, Loan_Enquiry, Loan_Application_Form, Salary_Slip_Verification, Vehicle_Photo_Check, Loan_Disbursement_Form, EMI_Payment_Tracking_Form, Customer_Feedback and Customer_Complaint—and one grid subform, Co_Applicant_Details. It includes 18 reports across multiple views, seven form workflows, four scheduled workflows, two approval processes and a three-stage Blueprint named Loan Disbursement Process.

The system supports validation of loan amount and interest rate, applicant onboarding emails, EMI status classification, disbursement-date autofill, overdue marking and advance EMI reminders. AI capabilities include OCR for salary slips, object detection for vehicle photographs and sentiment analysis for customer feedback and complaints. Two dashboards provide management-level visibility. Role-based access is configured through a CEO role and Administrator, Developer, Write, Read, Customer and Loan Applicant profiles.

The project aims to centralise loan-related information, improve data consistency, automate repetitive tasks and provide structured monitoring of the loan lifecycle. Detailed test evidence, deployment values and screenshots should be inserted from the actual application wherever indicated in this report.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Smart Bank Loan Management and Automation System
Application Domain	Banking, Financial Services and Loan Lifecycle Automation
Student Name and Register Number	HARINI NAGARAJAN – 240701167
Course Code and Title	CB23F35 – Low-Code Application Development using Zoho Creator
Course Faculty	Dr. Jinu Sophia
Platform	Zoho Creator
Application Link Name	loan-application
Published / Live URL	https://creatorapp.zoho.in/harininagarajan.2024.cse1592/loan-application
Forms and Subforms	10 forms + 1 grid subform (Co_Applicant_Details)
Reports	18 reports in multiple views
Pages / Dashboards	2 – Bank Loan Application Command Center and Loan_Management_Dashboard
Workflows	7 form workflows, 4 scheduled workflows, 2 approval processes and 1 Blueprint
Deluge Functions / Custom Actions	6 documented scripts (DS-01 to DS-06)
Integrations	No external core-banking or credit-bureau API integration is recorded
AI Features	OCR, Object Detection and Sentiment Analysis
Role and Profiles	CEO role; Administrator, Developer, Write, Read, Customer and Loan Applicant profiles
Environments	Development (Stage) to Production; deployment date to be confirmed

TABLE OF CONTENTS

ABSTRACT ..... **iv**
PROJECT AT A GLANCE ..... **v**
LIST OF TABLES ..... **viii**
LIST OF FIGURES ..... **ix**
LIST OF ABBREVIATIONS ..... **x**
INTRODUCTION **1**
1.1 Background 1
1.2 Problem Statement 1
1.3 Objectives 2
1.4 Scope of the Project 2
1.5 Justification for a Low-Code Approach 2
1.6 Organisation of the Report 3
LITERATURE SURVEY AND SYSTEM STUDY **4**
2.1 Study of the Existing System 4
2.2 Limitations of the Existing System 4
2.3 Low-Code versus Traditional Development 4
2.4 Literature Survey 5
2.5 Emerging Trends in Low-Code Development 5
2.6 Proposed System 5
REQUIREMENT ANALYSIS **6**
3.1 Stakeholders and User Roles 6
3.2 Functional Requirements 6
3.3 Non-Functional Requirements 6
3.4 Hardware and Software Requirements 7
3.5 Use Case Model 7
SYSTEM DESIGN **8**
4.1 System Architecture 8
4.2 Data Model 8
4.3 Form Design 8
4.4 Process and Workflow Design 9
4.5 User Interface and Navigation Design 10
4.6 Access Control Design 10
IMPLEMENTATION **11**
5.1 Application Setup 11
5.2 Forms and Data Capture 11
5.3 Reports 11
5.4 Pages and Dashboards 12
5.5 Deluge Scripting 12
5.6 Workflows and Automation 12
5.7 User Management and Role-Based Access 13
5.8 AI and Analytics Features 13
TESTING AND DEBUGGING **14**
6.1 Testing Strategy 14
6.2 Test Cases 14
6.3 Debugging Deluge Scripts 14
6.4 Defect Log 14
6.5 User Acceptance Testing 15
DEPLOYMENT, SECURITY AND GOVERNANCE **16**
7.1 Environments and Deployment Strategy 16
7.2 Performance 16
7.3 Security Features 16
7.4 Governance and Administration 16
RESULTS AND DISCUSSION **17**
8.1 Application Output 17
8.2 Achievement of Objectives 17
8.3 Comparison with the Existing System 17
8.4 Limitations of the Low-Code Approach 17
8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum 18
CONCLUSION AND FUTURE ENHANCEMENTS **19**
9.1 Conclusion 19
9.2 Future Enhancements 19
9.3 Learning Outcomes and Career Relevance 19
REFERENCES **21**
APPENDIX A – FEATURE COVERAGE CHECKLIST **22**
APPENDIX B – COURSE OUTCOME MAPPING **23**
APPENDIX C – DELUGE SOURCE CODE **24**

APPENDIX D – USER MANUAL **26**
APPENDIX E – EVALUATION NOTES **27**
APPENDIX F – VIVA-VOCE RECORD SHEET **28**

LIST OF TABLES

- Table 2.1 Comparison of Low-Code and Traditional Development
- Table 2.2 Summary of Literature Survey
- Table 3.1 Stakeholders and User Roles
- Table 3.2 Functional Requirements
- Table 3.3 Non-Functional Requirements
- Table 3.4 Hardware and Software Requirements
- Table 3.5 Use Case Description – Submit Loan Application
- Tables 4.1–4.10 Form and Subform Specifications
- Table 4.11 Role–Permission Matrix
- Table 5.1 Reports Implemented
- Table 5.2 Deluge Scripts Implemented
- Table 5.3 Workflows Implemented
- Table 6.1 Test Cases
- Table 6.2 Defect Log
- Table 6.3 User Acceptance Testing Record
- Table 7.1 Deployment Checklist
- Table 7.2 Security Controls
- Table 8.1 Mapping of Objectives to Outcomes
- Table 8.2 Existing System versus Proposed Application
- Table A.1 Feature Coverage Checklist

LIST OF FIGURES

- Figure 2.1 As-Is Process Flow of the Existing Loan Processing System
- Figure 3.1 Use Case Diagram
- Figure 4.1 System Architecture
- Figure 4.2 Entity–Relationship Diagram
- Figure 4.3 Loan Lifecycle Workflow
- Figure 4.4 Navigation Structure and Command Center Wireframe
- Figure 5.1 Application Home and Navigation
- Figure 5.2 Loan_Application_Form in Form Builder
- Figure 5.3 Loan_Application_Form in Live Mode
- Figure 5.4 Loan_Status_Kanban Report
- Figure 5.5 Bank Loan Application Command Center
- Figure 5.6 Workflow Configuration
- Figure 5.7 Loan Disbursement Blueprint
- Figure 5.8 Roles and Permission Profiles
- Figure 5.9 AI Features in Use
- Figure 7.1 Development and Production Environments
- Figures 8.1–8.2 Actual application test scenarios

LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CEO	Chief Executive Officer
CRUD	Create, Read, Update, Delete
DS	Deluge Script
EMI	Equated Monthly Instalment
ER	Entity–Relationship
FR	Functional Requirement
IFSC	Indian Financial System Code
KPI	Key Performance Indicator
KYC	Know Your Customer
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
NPA	Non-Performing Asset
OCR	Optical Character Recognition
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UI	User Interface
ZML	Zoho Markup Language
Zia	Zoho AI assistant

CHAPTER 1: INTRODUCTION

1.1 Background

Commercial banks provide personal, home, vehicle and education loans to individuals and businesses. The lending process involves collecting borrower information, recording loan requirements, verifying identity and income documents, assessing creditworthiness, obtaining approval, releasing funds and monitoring Equated Monthly Instalment (EMI) repayments.

These activities involve several teams and large volumes of records. A centralised digital application can help staff access related information, monitor each application stage and reduce repetitive administrative work. This project uses Zoho Creator to organise the loan lifecycle through forms, reports, workflows, approvals, a Blueprint and dashboards.

1.2 Problem Statement

Conventional loan processing may rely on paper files, spreadsheets and separate communication channels. Such methods make it difficult to maintain a complete borrower record and monitor the progress of each loan. Incorrect financial values may be entered, document verification can become delayed, approvals may lack a controlled sequence and overdue instalments may be identified late.

The main problems addressed are:

- Applicant details, KYC proofs, income documents and credit-check information may be stored in separate records.
- Manual spreadsheets may accept invalid loan amounts, tenures or interest rates unless validation is applied.
- Approval requests exchanged informally through email can delay processing and reduce visibility.
- Manual inspection of salary slips and vehicle collateral can create verification backlogs.
- Funds should not be released before the required verification and approval stages are completed.
- Late identification of unpaid instalments can affect recovery follow-up and increase credit risk.

1.3 Objectives

- Create forms and a related subform to capture enquiry, applicant, branch, loan application, verification, disbursement, repayment, feedback and complaint details.
- Validate loan amount and interest rate before accepting a loan application.
- Automate applicant onboarding communication and relevant status updates through workflows.
- Route loan applications and EMI records through approval processes.
- Control loan disbursement using a three-stage Blueprint.
- Schedule overdue EMI marking and advance payment reminders.
- Use OCR, object detection and sentiment analysis for selected document and customer-data tasks.
- Provide reports, dashboards and role-based access for operational staff and management.

1.4 Scope of the Project

In Scope: Applicant and branch record management; loan enquiries and multi-section applications; co-applicant details; salary-slip OCR; vehicle-photo object detection; application approval; Blueprint-controlled disbursement; EMI payment tracking; customer feedback and complaint records; automated email notifications; scheduled reminders; reports; dashboards; and access control within Zoho Creator.

Out of Scope: The source project report records no external core-banking integration or live credit-bureau API integration. External fund-transfer rails, live CIBIL pulls and statutory tax-ledger synchronisation are therefore treated as future enhancements, not implemented capabilities.

1.5 Justification for a Low-Code Approach

Low-code platforms provide visual tools for building forms, reports, workflows and user interfaces with limited conventional programming. Zoho Creator supports data capture, Deluge scripting, approval workflows, Blueprints, AI fields, role-based permissions and cloud hosting. This approach is suitable for a workflow-oriented application because it allows the developer to configure business processes and automate routine tasks without building the complete infrastructure from the beginning.

1.6 Organisation of the Report

Chapter 2 studies the existing process, its limitations and the low-code approach. Chapter 3 presents stakeholders and requirements. Chapter 4 describes the architecture, data model, forms, workflows and access control. Chapter 5 explains implementation. Chapter 6 presents the testing approach and test records. Chapter 7 discusses deployment, security and governance. Chapter 8 evaluates the application outputs and limitations. Chapter 9 concludes the report and identifies future enhancements.

CHAPTER 2: LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a conventional branch-based process, a customer enquiry is first recorded in a register or spreadsheet. Personal details, KYC documents and income proofs are collected separately. Staff inspect the documents and collateral, credit assessments are prepared, and approval requests are exchanged through email. After approval, funds are disbursed and repayment follow-up is performed by checking separate records.

Because each activity may use a different document or communication channel, staff need to combine information manually to understand the current position of a borrower.

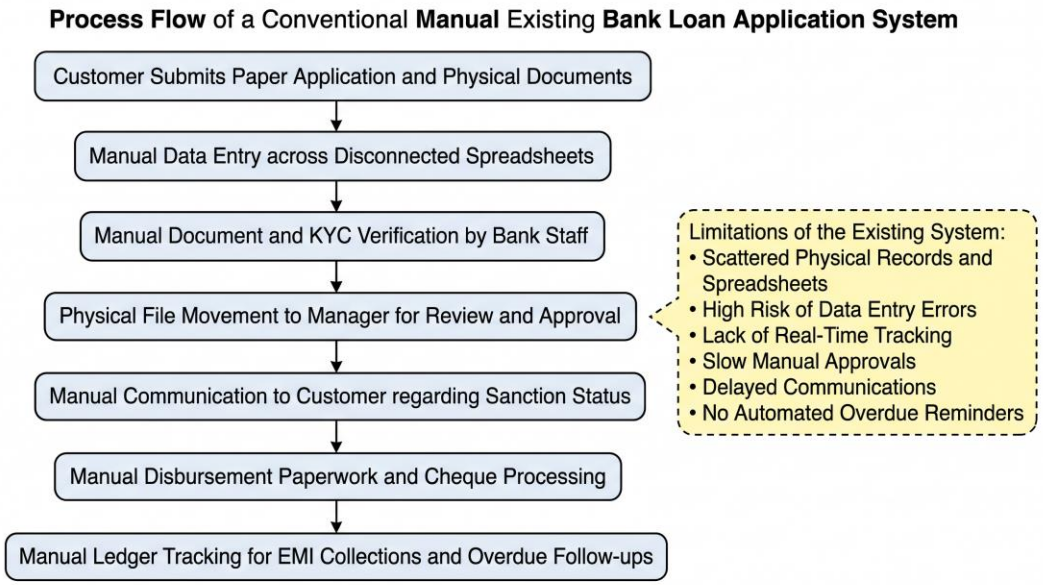


Figure 2.1: As Process Flow of the Existing Loan Application Management System

2.2 Limitations of the Existing System

- Borrower information is scattered across files, making a complete customer view difficult.
- Data-entry errors may remain undetected when validation rules are absent.
- Email-based approvals can be slow and may not provide a clear workflow status.
- Manual document and collateral inspection can delay verification.
- Disbursement without a defined sequence can create operational risk.
- Overdue instalments may be identified only during periodic reconciliation.
- Unstructured file access may expose records to users who do not need them.

2.3 Low-Code versus Traditional Development

Criterion	Traditional Development	Low-Code on Zoho Creator
Development time	Requires separate database, backend and interface engineering.	Forms, reports and workflows are configured visually.
Skills	Requires programming, database and deployment skills.	Requires process knowledge and platform configuration; Deluge supports custom logic.
Hosting	Servers and cloud infrastructure must be provisioned and maintained.	Application is hosted on the Zoho platform.
Mobile access	A separate mobile application may be required.	Zoho Creator provides mobile access.
Security	Authentication and permission logic must be implemented.	Built-in roles and permission profiles are available.
Maintenance	Code changes, testing and redeployment are required.	Many changes can be made in the visual builder.
Customisation	Offers broad flexibility.	Customisation is bounded by platform capabilities.
Cost	Can involve higher development and maintenance effort.	Can reduce development effort; subscription costs still apply.

2.4 Literature Survey

Source	Title / Resource	Key Contribution	Relevance
W. Rosam and O. Kalnichevski [1]	Low-Code Development: A Practical Guide	Explains low-code concepts and practical application development.	Conceptual basis for using low-code.
Zoho Corporation [2]	Zoho Creator Help Documentation	Describes forms, reports, workflows, approvals, Blueprints, AI fields and access.	Primary platform reference.
Zoho Corporation [3]	Deluge Script Reference Guide	Explains Deluge syntax, tasks and events.	Supports validation and scheduled scripts.
A. Sahay et al. [4], 2020	Supporting the understanding and comparison of low-code development platforms	Compares low-code platforms through a conceptual framework.	Supports platform selection.
D. Di Ruscio et al. [5], 2022	Low-code development and model-driven engineering: Two sides of the same coin?	Discusses the relationship between low-code and model-driven engineering.	Supports data-model-first design.
Gartner, Inc. [6], 2023	Magic Quadrant for Enterprise Low-Code Application Platforms	Reviews enterprise low-code platform vendors.	Provides industry context.

2.5 Emerging Trends in Low-Code Development

Citizen development, hyper-automation, AI-assisted application building and composable applications are relevant trends in low-code development. The project applies these ideas through visual form and report configuration, scheduled automation and embedded AI fields for selected verification and feedback tasks.

2.6 Proposed System

The proposed system replaces fragmented paper and spreadsheet records with a central Zoho Creator application. It organises loan data through related forms, applies validation before submission, supports AI-assisted extraction and classification, routes selected records for approval, controls disbursement through a Blueprint, schedules EMI reminders and overdue checks, and presents information through reports and dashboards.

CHAPTER 3: REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Role	Description	Responsibilities
Administrator	Application owner and developer.	Configure forms, Deluge logic, roles, permissions and security.
CEO	Top-level role in the role hierarchy.	Monitor lending volume, approval rates and branch performance.
Loan Officer	Branch staff using a Write profile.	Enter applicant information, review documents, monitor disbursement and repayment.
Credit Approver	Level-1 approver.	Review applications, approve or reject records and move disbursement through the Blueprint.
Loan Applicant	Customer / portal user.	Submit applications, upload supporting information and view permitted application status.

3.2 Functional Requirements

ID	Requirement	Target Module	Priority
FR-01	Capture applicant profile, salary, credit score and feedback sentiment.	Applicant_Details	High
FR-02	Register branches with IFSC codes, managers and addresses.	Branches	Medium
FR-03	Process multi-section loan applications with co-applicant grid and signature.	Loan_Application_Form	High
FR-04	Validate loan amount greater than zero and interest rate between 0% and 30%.	Loan_Application_Form	High
FR-05	Send applicant onboarding email with Applicant_ID.	Applicant_Details	Medium
FR-06	Extract text from uploaded salary slips using OCR.	Salary_Slip_Verification	Medium
FR-07	Detect objects in vehicle collateral photographs.	Vehicle_Photo_Check	Medium
FR-08	Route loan applications through Level-1 approval with notifications.	Loan_Application_Form	High
FR-09	Enforce a three-stage disbursement Blueprint.	Loan_Disbursement_Form	High
FR-10	Set disbursement date when status changes to Disbursed.	Loan_Disbursement_Form	High
FR-11	Classify EMI status as Paid, Pending or Overdue.	EMI_Payment_Tracking_Form	High
FR-12	Mark past-due EMI records as Overdue through a daily schedule.	Scheduled Workflow	High
FR-13	Send advance alerts for EMIs due within three days.	Scheduled Workflow	High
FR-14	Provide a management command center with KPI panels and charts.	Pages / Dashboards	High
FR-15	Analyse feedback and complaint tone using AI sentiment fields.	Customer_Feedback / Customer_Complaint	Low
FR-16	Restrict access through roles and permission profiles.	User Management	High

3.3 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Reports and dashboards should load promptly; the source report states a target of five seconds for expected record volumes.
NFR-02	Security	Users should access only data and actions permitted by their role; personally identifiable fields should be restricted.
NFR-03	Usability	The application should support form-based use through a standard browser and Zoho Creator mobile app.
NFR-04	Availability	The application is hosted on Zoho cloud and is accessible when an internet connection is available.
NFR-05	Maintainability	Forms, workflows and permissions should be modifiable through the visual builder.
NFR-06	Auditability	Record changes, approvals and workflow executions should be traceable through the available audit trail.

3.4 Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (edition to be confirmed from the account)
Scripting language	Deluge
Browser	Latest Google Chrome, Microsoft Edge or Mozilla Firefox
Mail service	Zoho Creator sendmail task / Zoho Mail delivery
Mobile	Zoho Creator mobile app for Android or iOS
Client hardware	Intel Core i3/i5 or above, 8 GB RAM and internet connection
API testing tool	Not specified in the source project report

3.5 Use Case Model

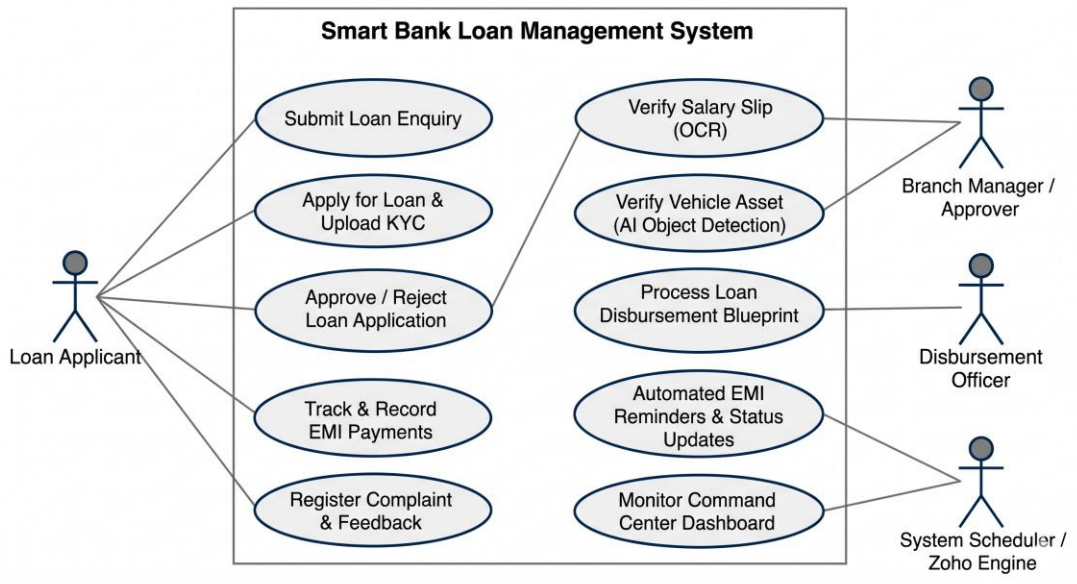


Figure 3.1 Use Case Diagram of the Application

Item	Description
Use Case ID	UC-01
Use Case Name	Submit Loan Application
Actors	Loan Officer or Loan Applicant; System performs validation and approval routing.
Pre-condition	The user is logged in with a profile that permits adding Loan_Application_Form records.
Main Flow	Enter loan type, amount, interest rate, address, co-applicant details and signature; submit the form; save the record and route it for Level-1 approval.
Alternate Flow	If the loan amount is zero or negative, or the interest rate is outside 0–30%, display an alert and cancel submission.
Post-condition	The record appears in reports and an approval task is created for the Credit Approver.
Related Requirements	FR-03, FR-04 and FR-08

CHAPTER 4: SYSTEM DESIGN

4.1 System Architecture

The application follows a multi-tier cloud architecture. The Presentation Layer provides responsive web and mobile interfaces and specialised report views such as Kanban, Map, Calendar, Timeline and Pivot. The Business Logic Layer executes Deluge event scripts, approval workflows, the Blueprint and AI field operations. The Data Layer stores application records in Zoho Creator's cloud database. Role-based access controls the actions available to users, while email notifications are sent through built-in mail actions.

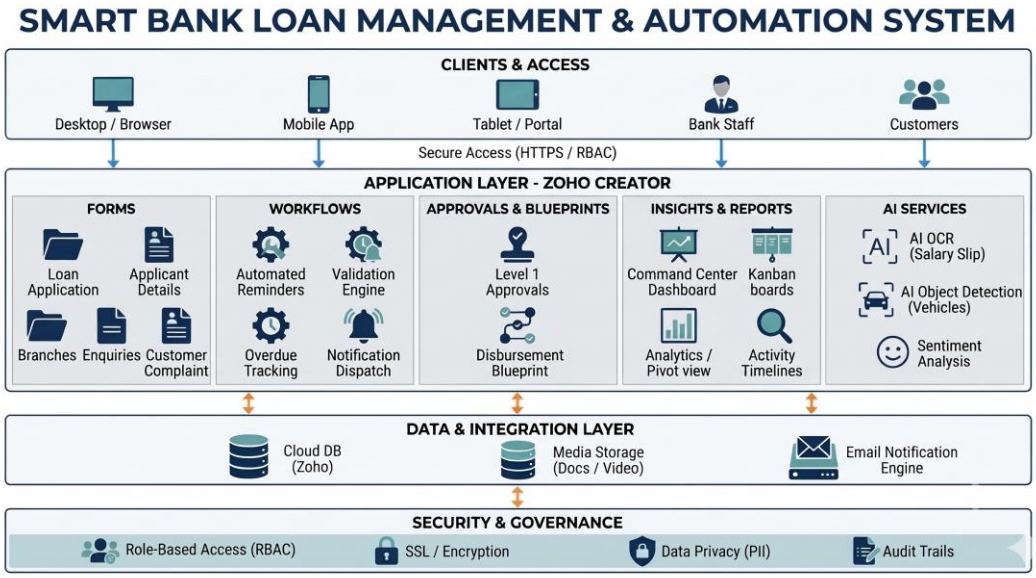


Figure 4.1 System Architecture of the Application

4.2 Data Model

The central Loan_Application_Form stores loan-specific information and references the Branches form through a lookup field. It contains the Co_Applicant_Details grid subform for joint applicants. Applicant_Details stores borrower profile and financial information. Salary_Slip_Verification and Vehicle_Photo_Check store verification records and AI outputs. Loan_Disbursement_Form and EMI_Payment_Tracking_Form support post-approval processing and repayment monitoring. The source report indicates that some related forms use Application_ID as a logical link; confirm lookup relationships in the live application before finalising the ER diagram.

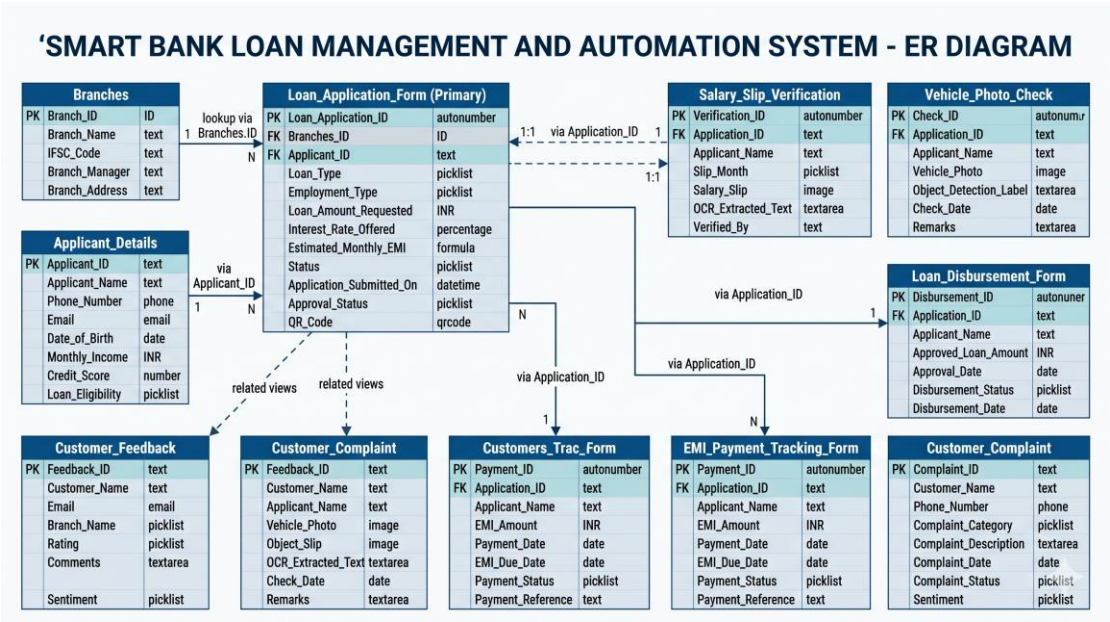


Figure 4.2 Entity-Relationship Diagram of the Application

4.3 Form Design

The application contains ten forms and one grid subform. The following summaries describe the fields identified in the project report. Mandatory settings should be checked in the live Form Builder because some exported field specifications are marked for confirmation.

Form	Purpose and key fields
Applicant_Details	Applicant_ID, Applicant_Name, Phone_Number, Email, Monthly_Income, Employment_Type, Credit_Score, Applicant_Feedback and AI Sentiment.
Branches	Branch_Name, IFSC_Code, Branch_Manager and Branch_Address.
Loan_Enquiry	Records customer loan enquiries and follow-up information; confirm the complete field list in the live form.
Loan_Application_Form	Loan_Application_ID, Branches lookup, Loan_Type, Applicant image, Residential_Address, Co_Applicant_Details, Property_Listing_URL, Loan_Amount_Requested, Interest_Rate_Offered, Applicant_Digital_Signature and QR_Code.
Salary_Slip_Verification	Verification_ID, Application_ID, Applicant_Name, Slip_Month, Salary_Slip, OCR output and Verified_By.
Vehicle_Photo_Check	Check_ID, Application_ID, Vehicle_Photo, Object_Detection output, Check_Date and Remarks.
Loan_Disbursement_Form	Captures loan disbursement details and status; includes the Blueprint-controlled process and disbursement date automation.
EMI_Payment_Tracking_Form	Tracks application reference, EMI amount, payment date, due date and Paid / Pending / Overdue status.
Customer_Feedback	Stores customer ratings and comments, with sentiment analysis.
Customer_Complaint	Stores complaint details and status, with sentiment analysis.

4.3.1 Co-Applicant Subform

Co_Applicant_Details is a grid subform within the loan application. It records Co_Applicant_Name, Relationship and Monthly_Income, allowing joint applicant information to be captured with the primary loan record.

4.3.2 Selected Field Specifications

Form / Field	Link Name	Type / Purpose
Applicant ID	Applicant_ID	Text / applicant reference
Applicant Name	Applicant_Name	Text / borrower identity
Phone Number	Phone_Number	Phone / contact information
Email	Email	Email / onboarding communication
Monthly Income	Monthly_Income	Currency (INR)
Employment Type	Employment_Type	Picklist / employment category
Credit Score	Credit_Score	Number / credit profile
Branch Name	Branch_Name	Text / branch directory
IFSC Code	IFSC_Code	Text / branch identifier
Loan Application ID	Loan_Application_ID	Auto Number / unique application ID
Loan Type	Loan_Type	Picklist / loan category
Loan Amount Requested	Loan_Amount_Requested	Currency (INR), validation greater than zero
Interest Rate Offered	Interest_Rate_Offered	Percent, validation 0–30%
Residential Address	Residential_Address	Address / used by map report
Applicant Signature	Applicant_Digital_Signature	Signature / consent capture
QR Code	QR_Code	QR Code / application reference
Salary Slip	Salary_Slip	Image / OCR input
Vehicle Photo	Vehicle_Photo	Image / object-detection input

4.4 Process and Workflow Design

The loan lifecycle begins with applicant registration and enquiry. The applicant or loan officer submits the loan application and supporting information. Validation checks the loan amount and interest rate. The application is routed for Level-1 approval. Verification records support salary-slip and vehicle-photo checks. Approved applications proceed to the Loan Disbursement Process Blueprint. Once the disbursement status reaches Disbursed, the date is populated. EMI records are then monitored through status classification and scheduled reminders.

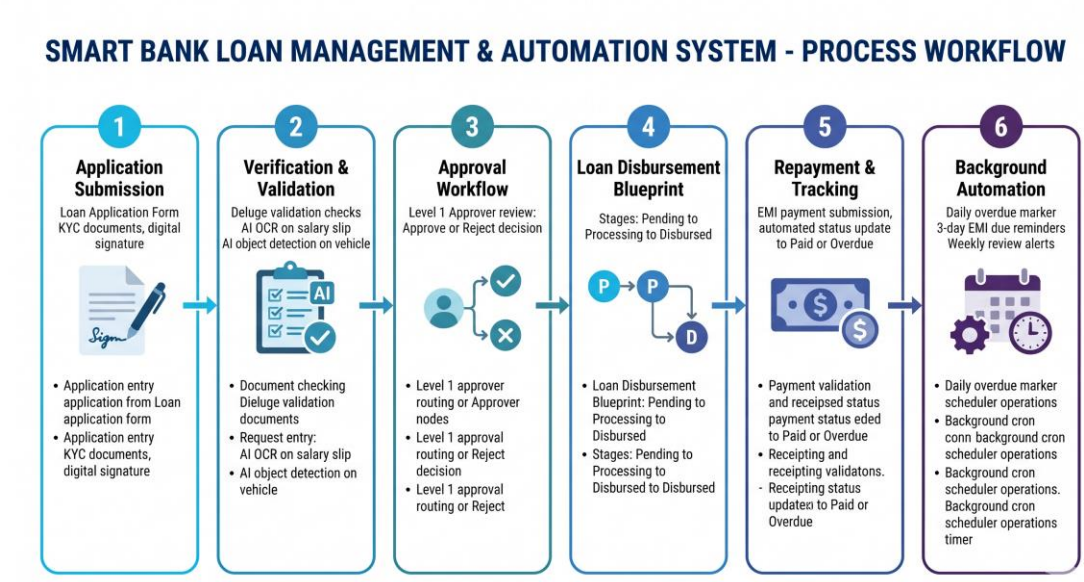


Figure 4.3 Workflow / Process Flow for the Loan Application Management System

4.5 User Interface and Navigation Design

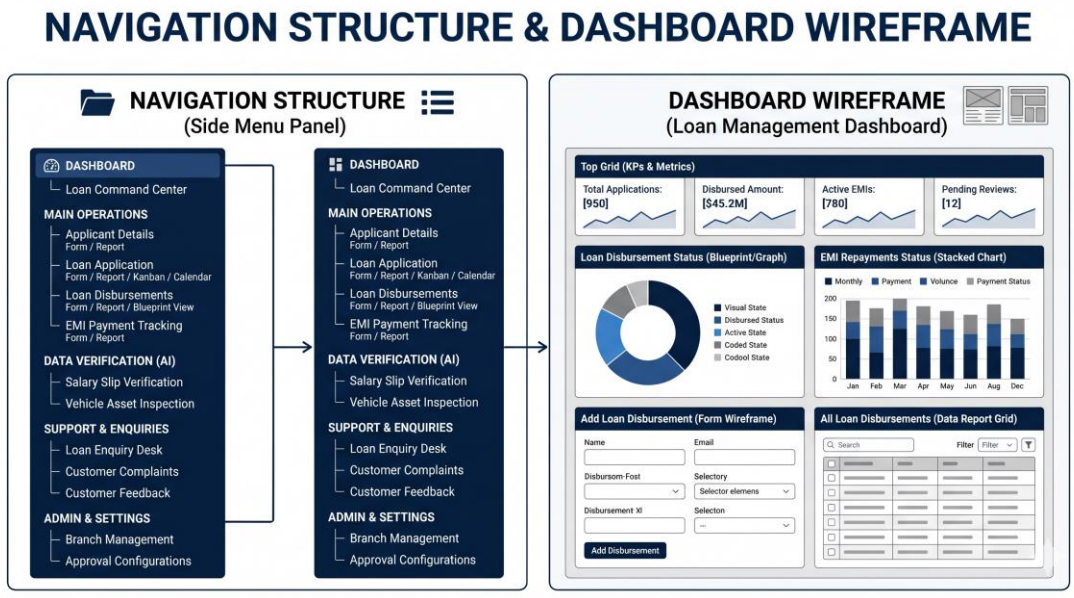


Figure 4.4: Navigation Structure and Wireframe of the Application and Loan Management Dashboard

The application provides form and report navigation for the loan lifecycle. Two pages are identified: Bank Loan Application Command Center and Loan_Management_Dashboard. The Command Center combines key panels, a disbursement status board, EMI chart and gauge, search, form and report components. Insert screenshots of the actual page builder and live application.

4.6 Access Control Design

The application defines a CEO role and the profiles Administrator, Developer, Write, Read, Customer and Loan Applicant. The project report identifies Administrator as the owner/developer profile and describes operational users such as Loan Officer and Credit Approver. Exact permissions should be documented from the live permission matrix.

Role / Profile	Intended access (verify in live settings)
Administrator	Application configuration and full administrative access.
CEO	Management-level visibility of lending and branch performance.
Developer	Application development and configuration according to assigned permissions.
Write	Create and update permitted operational records.
Read	View permitted records without editing.
Customer	Customer-facing access as configured.
Loan Applicant	Access to permitted application submission and status information.

CHAPTER 5: IMPLEMENTATION

5.1 Application Setup

The application was developed on the Zoho Creator low-code platform. The application link name recorded in the project report is loan-application. The development process includes configuring forms, reports, pages, Deluge scripts, workflows, approval processes, a Blueprint, AI fields and user permissions. Add the actual Zoho Creator edition, published URL and deployment date after verifying them in the account.

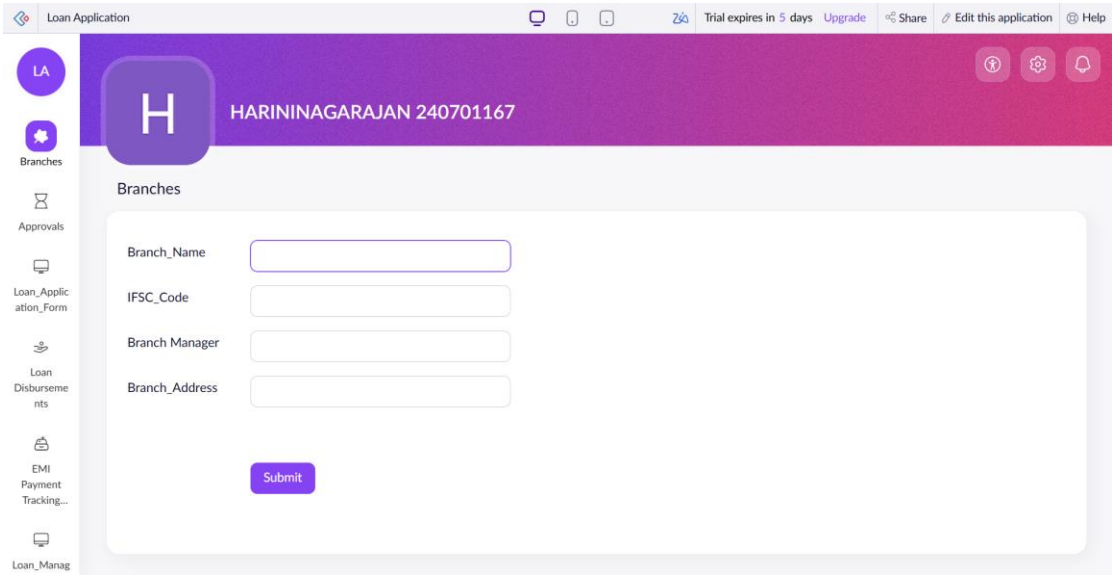


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

Ten forms organise the information required across the loan lifecycle. Applicant_Details and Branches maintain master information. Loan_Enquiry and Loan_Application_Form capture customer interest and formal loan requests. Co_Applicant_Details stores joint applicant information. Salary_Slip_Verification and Vehicle_Photo_Check support document and collateral review. Loan_Disbursement_Form and EMI_Payment_Tracking_Form manage fund release and repayments. Customer_Feedback and Customer_Complaint capture post-service information.

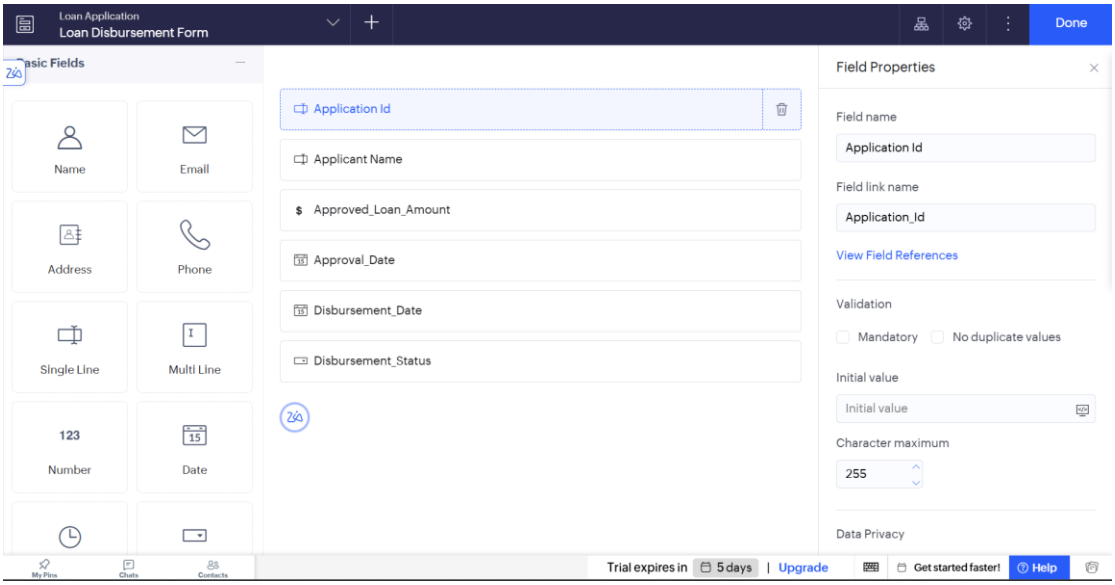


Figure 5.2: Loan Disbursement Form in Form Builder

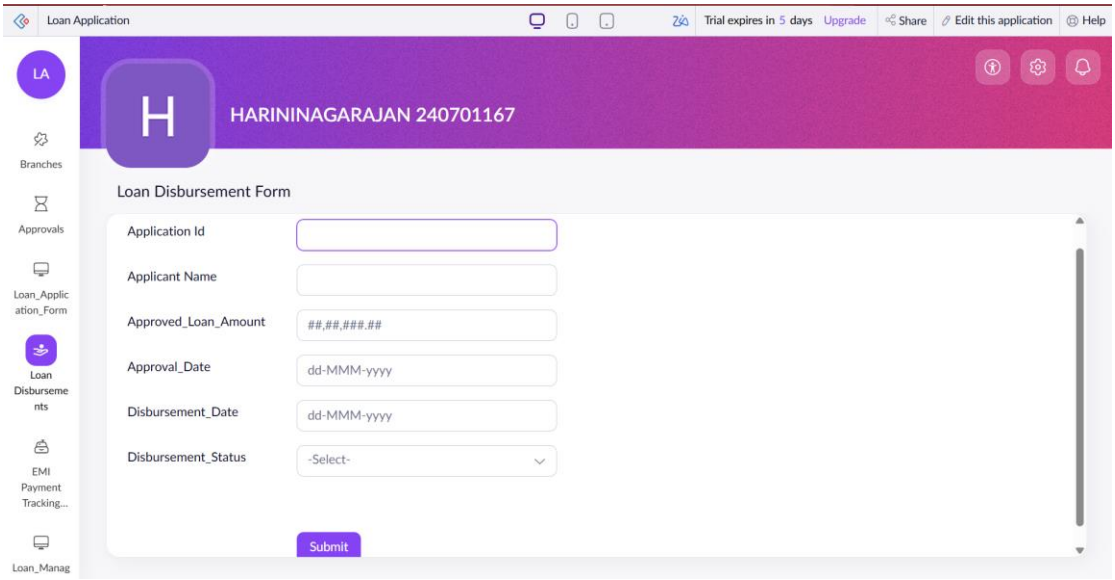


Figure 5.3 : Loan Disbursement Form in Live Mode

5.3 Reports

The application contains 18 reports across multiple views. The report views support record-level review, status tracking, time-based scheduling, geographic analysis and summary analytics.

Report / View	Purpose
All_Branches	View registered bank branches.
Loan_Application_Form_Report	View loan application information.
All_Loan_Applications_List	List loan application records.
Loan_Applications_Spreadsheet	Spreadsheet-style loan application view.
Loan_Status_Kanban	Track applications through status columns.
Callback_Schedule_Calendar	View applicant callback schedules.
Applicant_Residential_Map	Locate applicants by residential address.
Application_Activity_Timeline	View application activity over time.

All_Loan_Disbursements	Review disbursement records.
EMI_Payment_Tracking_Form_Report	Monitor EMI payment records.
Applicant_Details_Report	Review applicant profile and financial details.
Customer_Feedback_Report	Review customer feedback and ratings.
Customer_Complaint_Report	Monitor complaints and resolution information.
Salary_Slip_Verification_Report	Review salary-slip verification records.
Vehicle_Photo_Check_Report	Review vehicle photo inspection records.
Loan_Enquiry_Report	Track loan enquiries.
LoanType_By_EmploymentType_Pivo	Pivot analysis of loan type and employment type.
LoanType_Distribution_Chart	Chart-based loan type distribution.

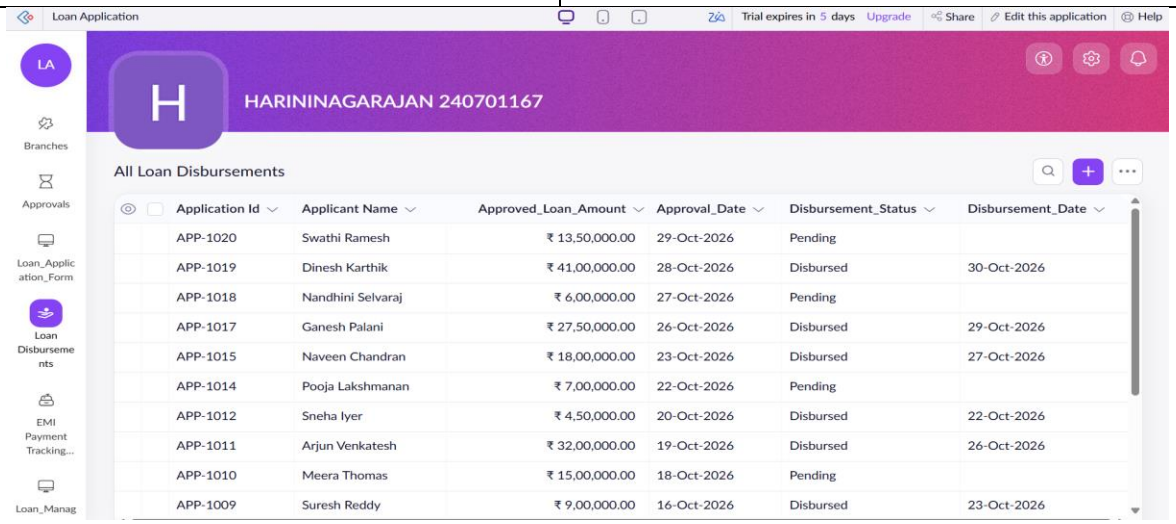


Figure 5.4: All Loan Disbursements Report

The project structure also identifies List, Spreadsheet, Kanban, Calendar, Map, Timeline, Pivot Table, Chart and Custom List / Summary-style views. Confirm the exact report type assigned to each report in the live application before submitting screenshots.

5.4 Pages and Dashboards

Two dashboards are recorded:

- Bank Loan Application Command Center: includes application-related KPI panels, a loan disbursement status board, EMI payment analysis chart, EMI gauge, disbursement form, search option, disbursement report and an action button.
- Loan_Management_Dashboard: includes a column chart showing total approved loan amount grouped by disbursement status.

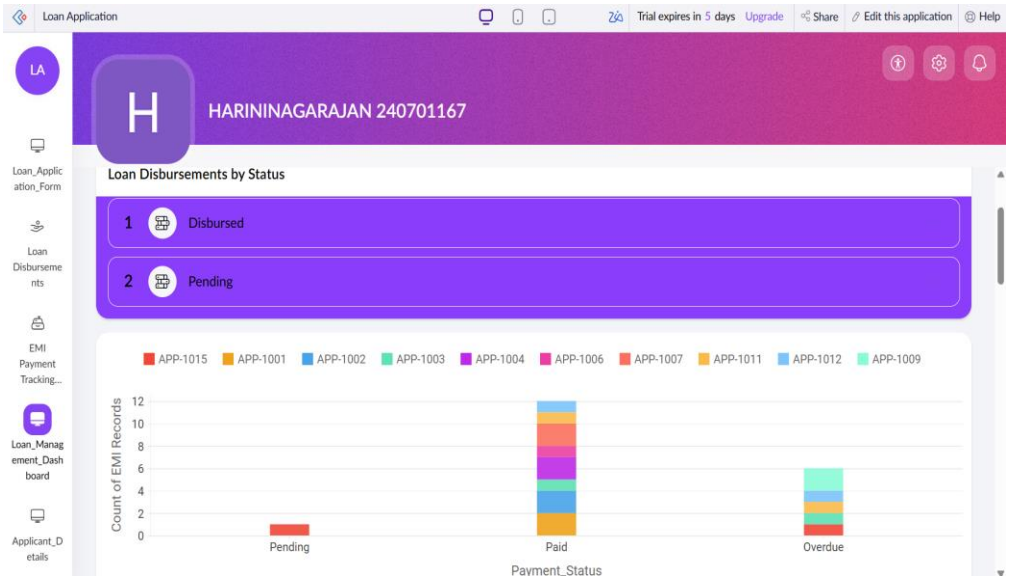


Figure 5.5: Loan Management Dashboard

5.5 Deluge Scripting

The project report records six documented Deluge scripts (DS-01 to DS-06). The scripts support validation and automation across the application. The available project summary specifically identifies loan amount and interest-rate validation, EMI status classification, disbursement date autofill and applicant onboarding email. Insert the complete verified scripts in Appendix C and map each script to its form event.

Script / Logic	Purpose
DS-01 – Loan validation	Reject a loan amount that is not greater than zero and an interest rate outside 0–30%.
EMI status logic	Classify an EMI record as Paid, Pending or Overdue using payment and due-date information.
Disbursement date autofill	Populate the disbursement date when the status changes to Disbursed.
Applicant onboarding email	Send a welcome email containing the Applicant_ID.
Other documented scripts	Add exact DS-02 to DS-06 names, trigger events and source code after checking the application export.

5.6 Workflows and Automation

The application records seven form workflows, four scheduled workflows, two approval processes and one Blueprint.

Workflow category	Count	Function
Form workflows	7	Validation, status classification, date autofill, onboarding email and other form-triggered actions.
Scheduled workflows	4	Daily overdue marking, advance EMI notices, weekly pending-review digest and birthday greetings.
Approval processes	2	Loan application approval and EMI payment review.
Blueprint	1	Three-stage Loan Disbursement Process.

Scheduled automation includes a daily job to mark past-due EMIs as Overdue, advance notifications for EMIs due within three days, a weekly reminder for pending loan reviews and birthday greetings. The exact trigger time, criteria and recipients should be confirmed from the workflow configuration screenshots.

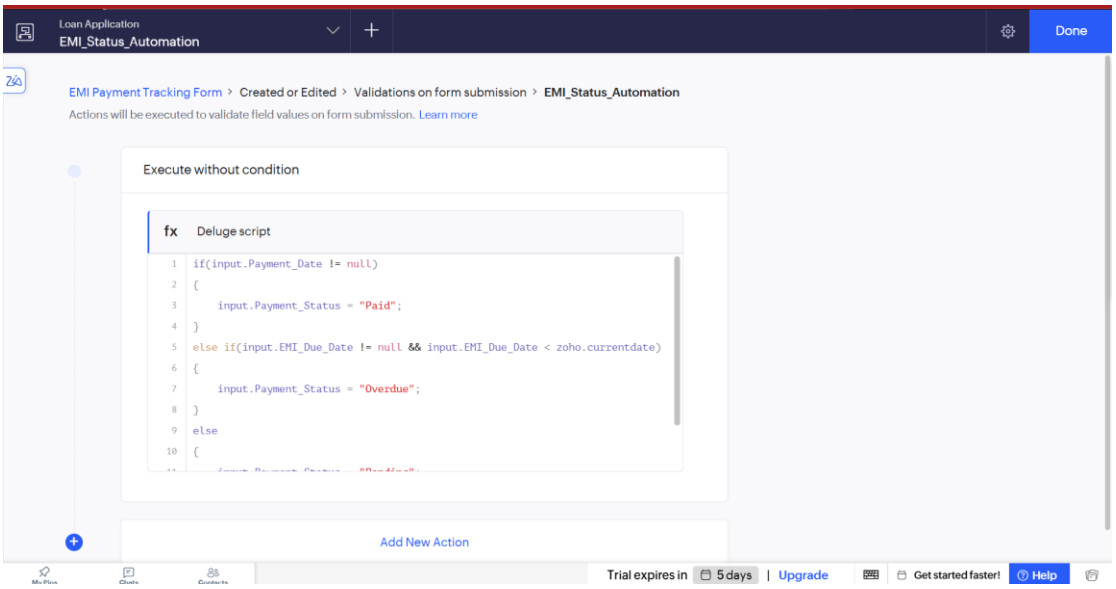


Figure 5.6: Workflow Configuration

5.7 User Management and Role-Based Access

The application uses a CEO role and six permission profiles: Administrator, Developer, Write, Read, Customer and Loan Applicant. Role-based access is intended to limit the actions available to each user. The submitted report should include a screenshot of the configured roles and permission profiles, with any personal email addresses or credentials hidden.

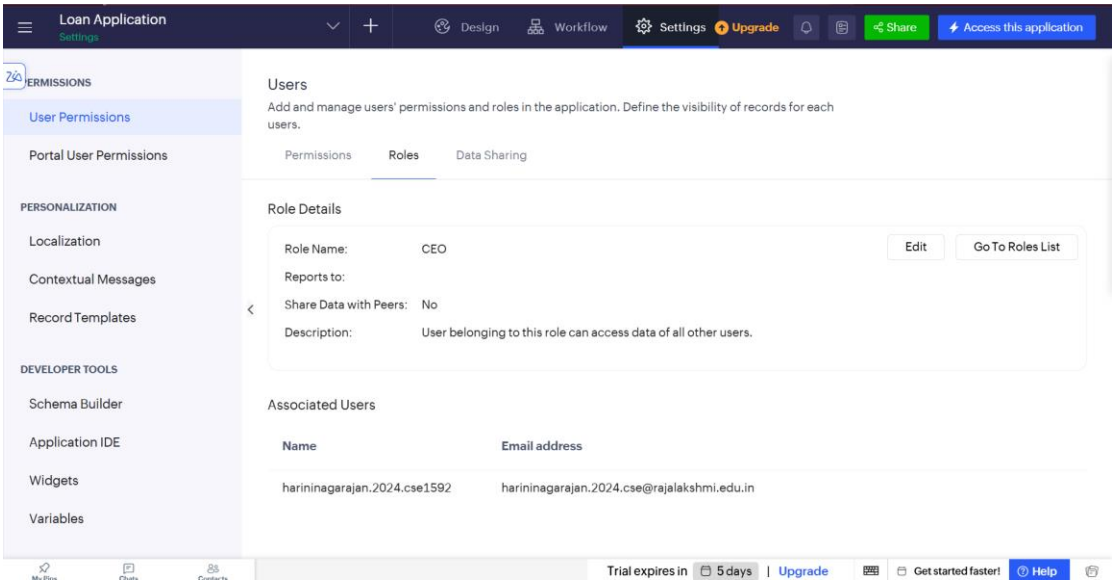


Figure 5.8: Roles and Permission Profiles Configuration

5.8 AI and Analytics Features

AI Feature	Input	Purpose
OCR	Uploaded salary slip in Salary_Slip_Verification.	Extract text from salary-slip images to support verification.
Object Detection	Vehicle photograph in Vehicle_Photo_Check.	Identify objects in the uploaded vehicle image for collateral review.
Sentiment Analysis	Feedback and complaint text.	Classify the tone of customer feedback and complaints.

These AI features assist staff; the report does not claim that they replace human verification or make final credit decisions.

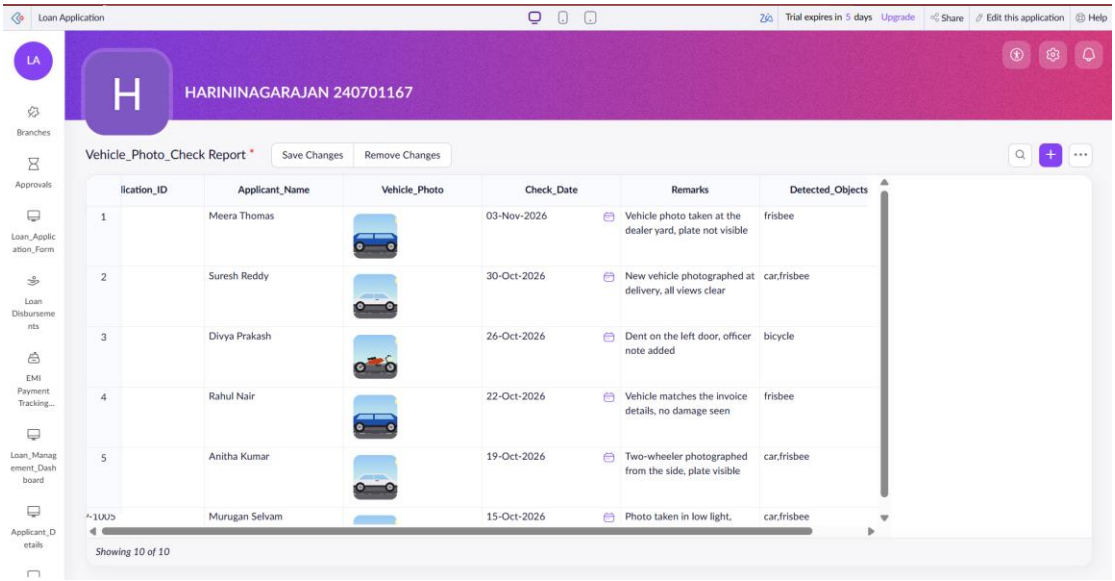


Figure 5.9: AI/Analytics Feature in Use

CHAPTER 6: TESTING AND DEBUGGING

6.1 Testing Strategy

Testing should verify the behaviour of each form, validation rule, report, workflow, approval process, Blueprint, scheduled task, AI field and permission profile. Test records should use non-sensitive sample data. Each test should record the input, expected result, actual result, status and evidence screenshot.

6.2 Test Cases

Test ID	Test Scenario	Expected Result	Actual Result / Status
TC-01	Submit applicant details with valid values.	Applicant record is saved and onboarding email is triggered where configured.	To be completed from live test.
TC-02	Submit a loan application with loan amount greater than zero and interest rate within 0–30%.	Application is accepted and routed for approval.	To be completed.
TC-03	Submit a loan application with zero or negative loan amount.	Validation alert is displayed and submission is cancelled.	To be completed.
TC-04	Enter an interest rate below 0% or above 30%.	Validation alert is displayed and submission is cancelled.	To be completed.
TC-05	Submit a salary slip for OCR processing.	OCR field returns extracted text where supported.	To be completed.
TC-06	Upload a vehicle photograph.	Object Detection field returns detected-object information where supported.	To be completed.
TC-07	Submit a loan application for approval.	Approval task is routed to the Level-1 approver.	To be completed.
TC-08	Move a disbursement record through the Blueprint.	The configured three-stage process controls the transitions.	To be completed.
TC-09	Change disbursement status to Disbursed.	Disbursement date is populated by the configured workflow.	To be completed.
TC-10	Create an EMI with a past due date and unpaid status.	Record is classified or scheduled for marking as Overdue.	To be completed.
TC-11	Create an EMI due within three days.	Advance reminder is sent according to schedule.	To be completed.
TC-12	Open reports and dashboards using permitted roles.	Only permitted records and actions are available.	To be completed.

6.3 Debugging Deluge Scripts

Deluge scripts should be debugged by checking field link names, data types, event triggers, criteria and execution results. For validation, test boundary values such as zero, negative amounts, 0% interest and 30% interest, along with values outside the allowed range. For date-based automation, test past, current and future due dates. Record any corrections made to scripts and retest the affected workflow.

6.4 Defect Log

Defect ID	Description	Severity	Correction	Status
D-01	Record any validation or field-mapping issue found during testing.	To be assigned	Document the actual correction.	To be completed.
D-02	Record any workflow, approval or Blueprint transition issue.	To be assigned	Document the actual correction.	To be completed.

D-03	Record any report, dashboard or permission issue.	To be assigned	Document the actual correction.	To be completed.

The rows above are a defect-log template, not confirmed defects. Remove rows that were not observed during actual testing.

6.5 User Acceptance Testing

User acceptance testing should be carried out using the configured user profiles, not only the administrator account. A Loan Officer should test record entry and review, a Credit Approver should test approval and Blueprint transitions, a Loan Applicant should test permitted submission and status access, and the CEO should review management dashboards. Insert actual feedback and signatures only after the tests are performed.

CHAPTER 7: DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The project summary records Development (Stage) and Production environments. The application should be validated in the development environment before publication. After deployment, confirm that forms, reports, workflows, approvals, Blueprint transitions, AI fields and permissions behave as expected in production.

Deployment Check	Evidence / Status
Application name and link name	loan-application – verify in live application.
Development environment	Insert screenshot of Stage environment.
Production publication	Insert screenshot of published application.
Forms and reports	Verify all required forms and reports are available.
Workflows and approvals	Verify triggers, recipients and approval routing.
Permissions	Verify role and profile access using test accounts.
Deployment date	Insert actual production deployment date.

7.2 Performance

The non-functional requirement specifies a target of five seconds for reports and dashboards under the expected record volume. This is a target stated in the project requirements, not a measured performance result. Record actual loading times under representative test conditions if performance claims are required.

7.3 Security Features

- Role and permission profiles restrict access to permitted actions.
- Read and Write profiles distinguish view-only access from record editing.
- Personally identifiable information such as phone numbers, email addresses and residential addresses should be restricted to authorised users.
- Administrative access should be limited to trusted application administrators.
- Demo credentials and personal account details should not be printed in the report.

7.4 Governance and Administration

The administrator is responsible for maintaining forms, workflows, scripts, permissions and application configuration. Changes should be tested before production publication. User access should be reviewed periodically, and workflow failures or data-quality issues should be investigated using the platform's available logs and audit information.

CHAPTER 8: RESULTS AND DISCUSSION

8.1 Application Output

The application structure provides a unified set of forms for applicant registration, branch management, loan enquiry, application submission, document verification, collateral checking, disbursement, EMI tracking, customer feedback and complaints. Reports present records through list, spreadsheet, Kanban, calendar, map, timeline, pivot and chart views. The two dashboards provide an operational and management-level view of selected loan information.

Insert actual screenshots for the following outputs: application home, loan application form, Loan_Status_Kanban, callback calendar, applicant map, disbursement form, EMI report, AI fields, approval screen, Blueprint and both dashboards.

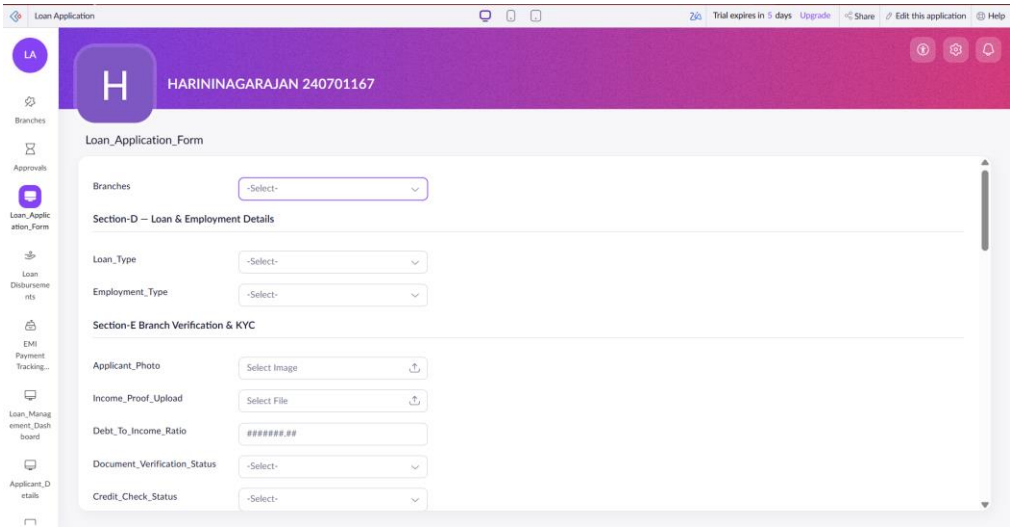


Figure 8.1: Scenario Step 1 – Recording Loan Application Form

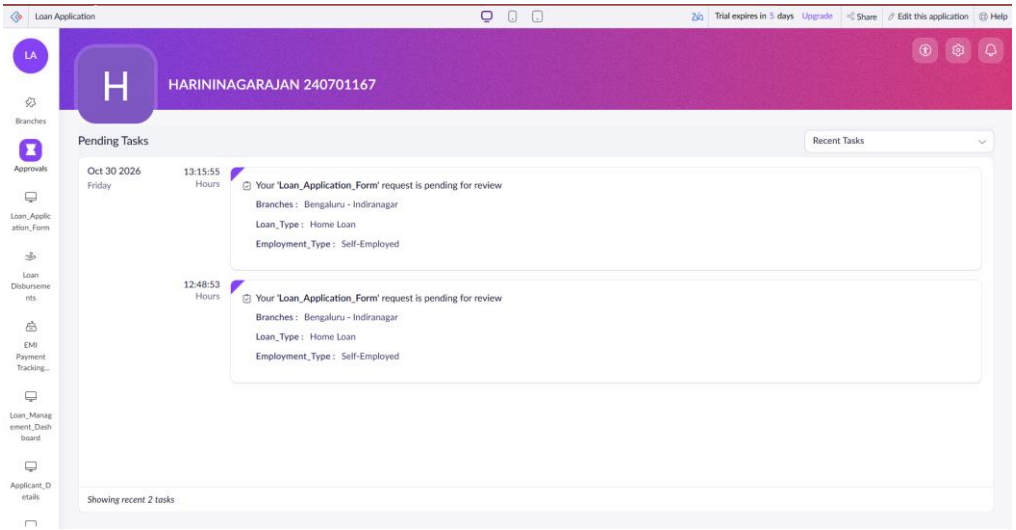


Figure 8.2: Scenario Step 2 – Approval Records Stored

8.2 Achievement of Objectives

Objective	Related Application Feature	Evidence to Attach
Centralise loan lifecycle data	Ten forms and one co-applicant subform.	Screenshots of forms and sample records.
Improve data accuracy	Loan amount and interest-rate validation.	Valid and invalid submission test evidence.
Automate communication	Applicant email and scheduled reminders.	Workflow configuration and email evidence.
Control approvals and disbursement	Two approval processes and Loan Disbursement Process Blueprint.	Approval and Blueprint screenshots.
Improve EMI monitoring	EMI classification, daily overdue job and advance alerts.	EMI records and scheduled workflow evidence.
Support verification	Salary-slip OCR and vehicle-photo object detection.	AI field screenshots and sample output.
Provide management visibility	18 reports and two dashboards.	Report and dashboard screenshots.
Restrict access	CEO role and six permission profiles.	Role and permission screenshots.

8.3 Comparison with the Existing System

Area	Existing Manual Process	Proposed Application
Record storage	Paper files and separate spreadsheets.	Centralised Zoho Creator forms.
Data validation	Manual checking; invalid values may be missed.	Configured validation for loan amount and interest rate.
Approval	Informal email-based routing.	Configured Level-1 approval processes.
Disbursement	Manual coordination.	Three-stage Blueprint.
Document review	Manual salary-slip and vehicle-photo inspection.	OCR and object detection fields assist review.
EMI monitoring	Periodic manual reconciliation.	Status classification and scheduled overdue marking.
Reminders	Manual follow-up.	Scheduled advance EMI and pending-review notifications.
Reporting	Manual compilation.	Multiple report views and dashboards.
Access	File-based access may be broad.	Role and permission profiles.

8.4 Limitations of the Low-Code Approach

The application depends on the features, limits and availability of the Zoho Creator platform. Complex banking integrations, live credit-bureau checks and external payment rails are not recorded as implemented. AI extraction and classification outputs require human review. Custom requirements beyond the platform's capabilities may require external APIs or pro-code extensions. Production performance and security must be verified under actual usage conditions.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Visual form, report, page and workflow builders provide no-code capabilities for configuring the basic application structure. Deluge scripts introduce low-code development for validation, date handling and automation. Future integration with core banking, credit bureaus or payment systems may require API development and pro-code components. The project therefore demonstrates how a low-code platform can combine visual configuration with targeted scripting.

CHAPTER 9: CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Smart Bank Loan Management and Automation System demonstrates the use of Zoho Creator to organise important activities in the retail loan lifecycle. It brings applicant, branch, enquiry, application, verification, disbursement, EMI, feedback and complaint information into a structured application. Reports and dashboards support monitoring, while Deluge workflows, approval processes and a Blueprint automate selected business steps.

The project also demonstrates the use of OCR, object detection and sentiment analysis as supporting AI capabilities. Role-based access provides a foundation for separating administrative, operational, management and applicant access. The application provides a low-code foundation for improving record organisation and reducing repetitive manual work. Final claims about accuracy, speed and user acceptance should be supported by actual test evidence.

9.2 Future Enhancements

- Integrate secure core-banking services and payment gateways for verified disbursement and repayment updates.
- Connect to authorised credit-bureau APIs for live credit checks, subject to regulatory and institutional approval.
- Add a customer portal for application tracking, document submission and EMI statements.
- Introduce more detailed loan eligibility rules and configurable risk scoring.
- Provide audit-ready reports, exception alerts and branch-level performance comparisons.
- Improve document verification with additional checks and human review queues.
- Add multilingual notifications and accessible mobile workflows.
- Strengthen monitoring through operational metrics, backup procedures and periodic access reviews.

9.3 Learning Outcomes and Career Relevance

The project provides practical experience in low-code application development, data modelling, form and report design, Deluge scripting, workflow automation, approval configuration, Blueprint design, AI field configuration and role-based access control.

REFERENCES

1. [1] W. Rosam and O. Kalnichevski, Low-Code Development: A Practical Guide.
2. [2] Zoho Corporation, Zoho Creator Help Documentation. https://www.zoho.com/creator/help/
3. [3] Zoho Corporation, Deluge Script Reference Guide. https://www.zoho.com/deluge/help/
4. [4] A. Sahay et al., “Supporting the understanding and comparison of low-code development platforms,” 2020.
5. [5] D. Di Ruscio et al., “Low-code development and model-driven engineering: Two sides of the same coin?”, 2022.
6. [6] Gartner, Inc., Magic Quadrant for Enterprise Low-Code Application Platforms, 2023.

APPENDIX A – FEATURE COVERAGE CHECKLIST

Feature	Included in project structure	Evidence
Ten application forms	Yes	Form list / screenshots
Co_Applicant_Details subform	Yes	Loan application form screenshot
18 reports	Yes	Report list / screenshots
Two dashboards	Yes	Dashboard screenshots
Seven form workflows	Yes	Workflow configuration
Four scheduled workflows	Yes	Schedule configuration
Two approval processes	Yes	Approval configuration
One disbursement Blueprint	Yes	Blueprint screenshot
OCR	Yes	Salary slip OCR output
Object Detection	Yes	Vehicle image output
Sentiment Analysis	Yes	Feedback / complaint output
Role-based profiles	Yes	Permission settings
External core-banking integration	Not recorded	Future enhancement
Live credit-bureau API	Not recorded	Future enhancement

APPENDIX B – COURSE OUTCOME MAPPING

Course Outcome	Relevant Report Sections	Evidence
CO-1	Chapters 1–3	Problem definition, requirements and stakeholder analysis.
CO-2	Chapter 4	Architecture, data model, forms and workflow design.
CO-3	Chapter 5	Zoho Creator implementation, Deluge and automation.
CO-4	Chapter 6	Testing strategy and test evidence.
CO-5	Chapters 7–9	Deployment, security, evaluation and future enhancements.

APPENDIX C – DELUGE SOURCE CODE

```
if(input.Email != null && input.Email != "")
{
	sendmail
	[
		from :zoho.adminuserid
		to :input.Email
		subject :"Welcome to the Loan Application Portal, " + input.Applicant_Name
		message :"Dear " + input.Applicant_Name + ",<br><br>Thank you for registering with us. Your applicant ID is <b>" + input.Applicant_ID + "</b>.<br><br>Our team will contact you shortly regarding your loan application.<br><br>Regards,<br>Loan Application Team"
	]
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

7. Open the published Zoho Creator application URL provided by the application owner.
8. Sign in using an authorised account. Do not share passwords or include them in the report.
9. Use the navigation menu to open the required form, report or dashboard.

D.2 Instructions for Loan Officer

10. Open Applicant_Details and enter the applicant's permitted personal and financial information.
11. Use Loan_Enquiry to record customer enquiries and follow-up details.
12. Open Loan_Application_Form and enter loan type, branch, amount, interest rate and other required details.
13. Add co-applicant information in Co_Applicant_Details when applicable.
14. Upload supporting documents and submit the record for the configured verification and approval process.
15. Use the reports to monitor application, disbursement and EMI status.

D.3 Instructions for Credit Approver

16. Open the pending approval task using the authorised approver account.
17. Review the application and supporting information.
18. Approve or reject the record using the configured approval action.
19. For disbursement, follow the available stages in the Loan Disbursement Process Blueprint.

D.4 Instructions for Loan Applicant

20. Sign in using the authorised applicant access method.
21. Submit the loan application and required information.
22. Upload permitted supporting documents.
23. View only the application and status information made available by the configured permissions.

APPENDIX E – EVALUATION NOTES

Student Name	HARINI NAGARAJAN	**Register No.**	240701167
Project Title	Smart Bank Loan Management System	Date	10.11.2026

Table E.1: Evaluation Rubric for the Individual Project

Sl No.	**Criterion**	**Indicators**	**Max. Marks**	**Marks Awarded**
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge	Correctness and	15	

	Scripting & Workflows	depth of scripts; form, schedule, approval, Blueprint, batch workflows		
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90-100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70-89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50-69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	<50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name:

Signature of External Examiner

Name:

APPENDIX F – VIVA-VOCE RECORD SHEET

Student Name	HARINI NAGARAJAN	Register No.	240701167
Department	Computer Science And Engineering	Date of Viva	10.11.2026
Application URL Verified	Yes/No	Demo Duration	----------min

Table F.1: Viva-Voce Questions and Responses

S.No.	Viva Question	Student Response	Marks
1			
2			
3			
4			
5			

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1-8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Internal Examiner: _____

External Examiner:_____